

Applications:



Early wildfire detection

- Detect small temperature increases **before** fires spread
- Integrated with **AI fire prediction** models for automated alerts



Industrial fire safety

- Detect **overheating machinery** by detecting IR before flames appear
- Integrated into smart safety systems to trigger **auto-shutdowns**



Military and surveillance

Detect enemy **heat signatures**, missile launches or thermal activity



Environmental monitoring

Can detect climate change by detecting **heat variations** in various ecosystems

Current challenges:



Scalability and cost

Current MEMS based sensors are **expensive** to produce



Environmental factors

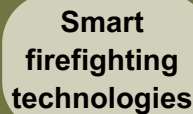
Harsh outdoor environments can **degrade** sensor performance



Interference from background sources

Refinement needed to **distinguish** wildfires from other heat sources

What does the future hold?



AI driven firefighting sensors can be used to **analyse** fire behaviour and **predict** fire paths



MEMS-based sensors in satellites can enable **global** wildfire tracking



Future missions could use these sensors to **detect** heat sources and **prevent** equipment malfunctions

CONCLUSION

The brochure highlights how the **biomimetic infrared sensing mechanism** of the *Melanophila acuminata* beetle can inspire the development of advanced **uncooled** IR sensors. The inspired sensor is effective, which is useful for early wildfire detection, military surveillance, and environmental monitoring.

CONTACT US



kj322@ic.ac.uk

CASE STUDY & WEBSITE



MAJOR REFERENCES

[1] Israelowitz, M., Kwon, J.-A., Rizvi, S.W.H., Gille, C. and von Schroeder, H.P. (2011). Mechanism of infrared detection and transduction by beetle *Melanophila Acuminata*. *Journal of Bionic Engineering*, 8(2), pp.129–139. doi:[https://doi.org/10.1016/s1672-6529\(11\)60018-8](https://doi.org/10.1016/s1672-6529(11)60018-8).

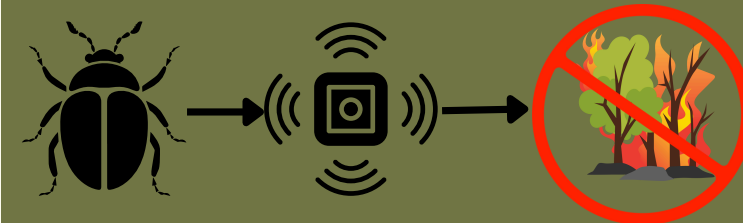
[2] Georg Siebke, Holik, P., Schmitz, S., Tätzner, S., Thiesler, J. and Siegfried Steltenkamp (2015). The development of a μ -biomimetic uncooled IR-Sensor inspired by the infrared receptors of *Melanophila acuminata*. *Bioinspiration & Biomimetics*, 10(2), pp.026007–026007. doi:<https://doi.org/10.1088/1748-3190/10/2/026007>.

[3] Klocke, D., Schmitz, A., Soltner, H., Bousack, H. and Schmitz, H. (2011). Infrared receptors in pyrophilous ('fire loving') insects as model for new un-cooled infrared sensors. *Beilstein Journal of Nanotechnology*, 2, pp.186–197. doi:<https://doi.org/10.3762/bjnano.2.22>.



BIOMIMETIC INFRARED SENSORS FOR FOREST FIRE DETECTION

*Inspired by the Melanophila
acuminata beetle*



Authors: Daniel Huang, Hiba Kadri, Nikita James, and Kangjun Ji (All authors contributed equally to this work)

INTRODUCTION

Forest fires can reach temperatures up to a **1000 degrees**, an environment typically unliveable for most organisms.

However, the black fire beetle, or *melanophila acuminata*, has been observed to approach these fires and use the lack of the burnt trees' defence mechanisms to **mate and lay their eggs**.

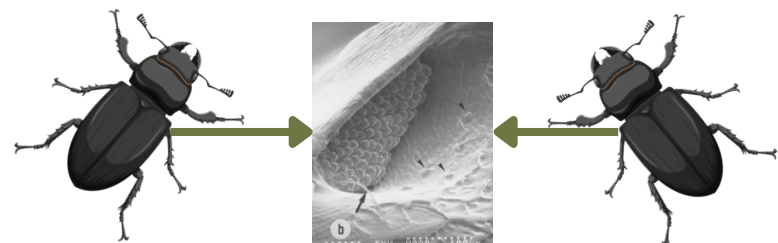


The mechanism by which these beetles find and approach these fires is through the detection of **infrared radiation**.

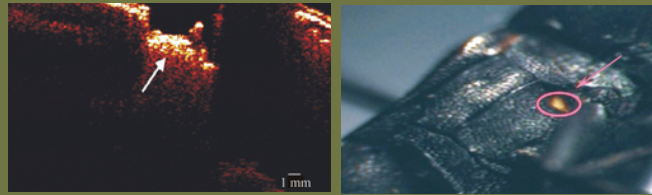
They have specialised **sensory pit organs** which convert the infrared radiation from the fires to heat, leading to a series of mechanical changes within the receptor, allowing **bioelectric nerve transmission**.

OBJECTIVE

To use this highly **sensitive** and extremely **rapid** response to infrared radiation as a basis for the development of invaluable sensors for human use.



METHODS



01 Infrared Absorption & Energy Conversion

The beetle's **infrared sensory organs** detect IR radiation (3 μm & 10 μm) via specialized sensilla. The cuticular dome absorbs IR, heating the liquid inside the microfluidic chamber to trigger detection.

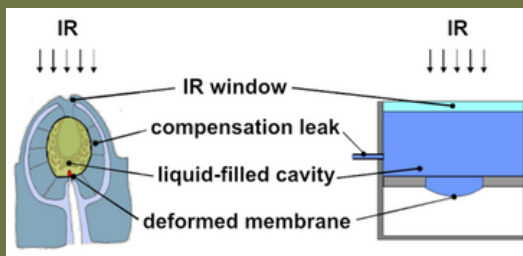
$$\Delta P = \beta \cdot \Delta T_{\text{mean}} - \frac{1}{\kappa \cdot V} \cdot \Delta V$$

02 Liquid Expansion & Pressure Increase

IR-induced heating **expands the liquid** in the sealed microfluidic core, but the rigid cuticular shell restricts it, increasing internal pressure, denoted in the equation above.

03 Membrane Deformation

Increased pressure deforms a flexible membrane, stretching mechanosensitive ion channels in the neuron's dendritic tip, similar to hair cell activation in the ear.



04 Neural Activation & Fire Detection

Membrane deformation opens ion channels (Na^+ , K^+ , Ca^{2+}), **depolarizing the neuron** and generating an action potential. The beetle processes the signal and navigates toward the thermal source, and hence toward the forest fire.

RESULTS

01 Sensory Organ Structures

Each sensory pit in the beetle contains **50–100 sensilla**. These sensilla are not just **thermal sensors**, they use a **protein-based photomechanical mechanism** for IR detection.

02 Performance of Different Materials

Cavity-Filling Material	Membrane Deflection (nm)
Water	~0.04 nm
Hydrocarbons (e.g., n-pentane, toluene)	~1.26 nm (10x increase)
Gas (CO_2)	~2.18 nm (Best performance)

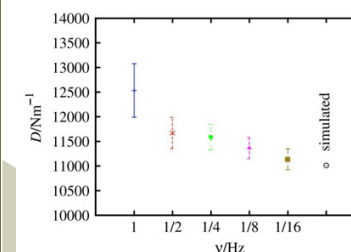
Gases such as **CO_2** significantly increase the deformation amplitude of the membrane, making it more suitable for biomimetic sensors than water.

03 An Uncooled MEMS-based IR Sensor

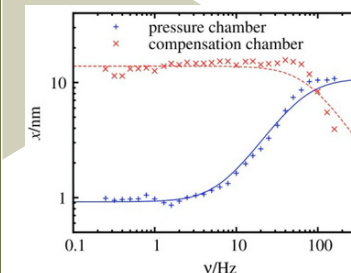
The design includes:

- A **pressure chamber** that mimics the sensillum's response to infrared-induced expansion.
- **Compensation mechanism** to minimize the impact of temperature fluctuations on sensitivity.
- **Bubble-free filling method** to enhance performance reliability.

The MEMS-based sensor achieved a sensitivity of $\sim 1 \text{ W/m}^2$, closely matching beetle behavioral experiments.



a) compares experimental and simulated spring constants, verifying that the **spring constant matches simulated expectations closely** (within $\sim 7\%$ difference).



b) for high frequencies, the membrane of the pressure chamber is deflected most, whereas the membrane of the compensation chamber is only deflected very little.